Cascade Speed–Current Control of a Separately-Excited DC Machine

Thyristor-rectifier drive: modelling, analytical PI/PID tuning, and Python/Simulink verification

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Abstract

This report presents the complete design and simulation of a cascade speed-and-current controller for a separately-excited DC machine supplied by a three-phase, six-pulse thyristor rectifier. Every part of the drive is developed in detail: the operation of the controlled rectifier and the synthesis of its firing angle, the averaged converter model used for control design, the plant transfer function and its Simulink realisation, and the analytical tuning of the current and speed controllers by pole–zero cancellation with a settling-time criterion. The design is verified twice over — a Python model that reproduces the tuning exactly and quantifies the parameter dependencies, and a hierarchy of MATLAB/Simulink models culminating in a full physical model with a detailed thyristor bridge. The studied operating point is an armature voltage of 220 V and a target speed of 3000 rpm.

Contents

1. System overview

1.1 The separately-excited DC machine

The plant is a separately-excited DC machine whose speed is regulated through its **armature voltage** u_a, the field excitation being held constant so that the flux term $k\Phi$ is a constant. The machine is then linear and is described by an electrical balance on the armature circuit and a mechanical balance on the rotor:

$$u_a(t) = R_a\, i_a(t) + (L_a + L_f)\frac{di_a}{dt} + k\Phi\,\omega(t), \qquad J\frac{d\omega}{dt} = k\Phi\, i_a(t) - (f_r + K_r)\,\omega(t),$$

where $k\Phi\,\omega$ is the back-EMF, $k\Phi\, i_a$ the electromagnetic torque, f_r the mechanical friction, and K_r a viscous load coefficient. Here L_f is the **smoothing inductor** placed in series with the armature to reduce the current ripple of the rectifier (its value, 276 mH, dominates the electrical time constant). The machine parameters (preset 1, confirmed against the Simscape *DC Machine* block) are listed in Table 1.

Table 1. Parameters of the studied DC machine.

Symbol	Quantity	Value
R_a	armature resistance	$2.33\ \Omega$
L_a	armature inductance	10.6 mH
L_f	series / smoothing inductance	276 mH
$k\Phi$	torque / back-EMF constant	0.57 V s/rad
J	rotor inertia	7.5×10^{-3} kg m^2
f_r	viscous friction	1.75×10^{-4} N m s
k	equivalent speed-loop damping (design)	1.25×10^{-2}

Eliminating the current between the two balances gives the armature-voltage-to-speed transfer function that underpins the whole study:

$$G(s) = \frac{\Omega(s)}{U_a(s)} = \frac{k\Phi}{(R_a + L_a s)(Js + f_r + K_r) + k\Phi^2}.$$

Its derivation and Simulink realisation are given in Section 3.

1.2 Cascade (nested-loop) control architecture

The controller uses a classical **cascade** structure with two nested loops. An **inner current loop** regulates the armature current i_a and is designed to be fast; its reference is produced by an **outer speed loop**, designed to be comparatively slow. A saturation block on the current reference gives natural **current limiting** during large transients such as start-up and load steps — essential for protecting the machine and the thyristors.

The design deliberately enforces a **time-scale separation**: the current loop is tuned to respond about ten times faster than the speed loop (response times 0.1 s and 1 s). Because the inner loop settles long before the outer loop reacts, the speed controller effectively sees the current loop as an ideal, instantaneous torque actuator, and the two loops can be designed independently (Section 4).

The complete signal chain is therefore

$$\omega_{\text{ref}} \to \boxed{\text{Speed PI}} \to i_{a,\text{ref}} \to \boxed{\text{Current PID}} \to U_{cm} \to \boxed{\text{Rectifier}} \to u_a \to \boxed{\text{DC machine}} \to \omega,$$

with i_a fed back to the inner summing junction and ω to the outer one.

2. The controlled rectifier and its model

The armature is supplied by a three-phase, **six-pulse thyristor rectifier** (a fully-controlled Graetz bridge). This section explains how the bridge works, how its firing angle is generated (the *Alpha generator*), and how the whole converter is condensed into the single transfer function $22/(T_c s+1)$ used for control design.

2.1 The six-pulse thyristor bridge

The bridge is the Simscape *Universal Bridge* configured with three arms and six **thyristors**, fed by a three-phase source of line-to-line voltage $U_{LL} = 220$ V (RMS) at $f = 50$ Hz. The six thyristors conduct in pairs, each for 120°, and there are **six commutations per mains period**, so the rectified voltage carries a ripple at

$$f_{\text{ripple}} = 6 \times 50 = 300 \text{ Hz},$$

which the 276 mH smoothing inductor filters into a nearly constant armature current. Unlike a diode bridge, a thyristor only conducts once its gate receives a pulse, so conduction can be *delayed* by a controllable **firing angle** α measured from the natural commutation instant. This delay is exactly the control handle of the drive.

2.2 Average output voltage and firing-angle control

Averaging the six-pulse waveform over one 60° segment gives the classical result for the mean DC output of a fully-controlled bridge:

$$V_d(\alpha) \;=\; \frac{3\sqrt{2}}{\pi}\,U_{LL}\,\cos\alpha \;=\; \frac{3\sqrt{6}}{\pi}\,V_\varphi\,\cos\alpha \;\approx\; 1.35\,U_{LL}\,\cos\alpha,$$

where $V_\varphi = U_{LL}/\sqrt{3}$ is the phase RMS voltage. With $U_{LL} = 220$ V the no-delay maximum is

$$V_{d0} = 1.35 \times 220 \approx 297 \text{ V}, \qquad V_d(\alpha) = 297\cos\alpha.$$

The converter is therefore **four-quadrant** in voltage: $\alpha < 90^\circ$ gives $V_d > 0$ (rectifier, motoring), $\alpha > 90^\circ$ gives $V_d < 0$ (inverter, allowing regenerative braking), and $\alpha = 90^\circ$ gives $V_d = 0$. The catch is that $V_d(\alpha)$ is **non-linear** (a cosine) in the control variable α — which is what the Alpha generator is there to fix.

2.3 The Alpha generator — a linearising firing law

The *Alpha generator* is the user-built subsystem that converts the current controller's output U_{cm} into the firing angle α (in degrees) fed to the six-pulse pulse generator. Its structure is shown in Figure 1.

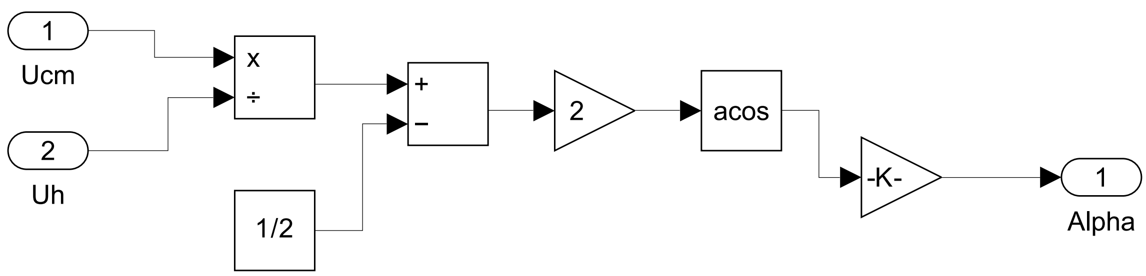


Figure 1. The Alpha generator subsystem: it produces $\alpha = \arccos(2U_{cm}/U_h - 1)$, the inverse-cosine firing law that linearises the thyristor bridge.

Reading it left to right, with $U_h = 10$ the reference amplitude:

$$x = \frac{U_{cm}}{U_h} \xrightarrow{-\frac{1}{2},\,\times 2} \left(\tfrac{2U_{cm}}{U_h} - 1\right) \xrightarrow{\text{arccos}} \alpha_{\text{rad}} \xrightarrow{\times\frac{180}{\pi}} \alpha_{\text{deg}}.$$

That is, $\boxed{\alpha = \arccos(2U_{cm}/U_h - 1)}$. The reason for the inverse cosine is elegant: substituting it into the bridge law cancels the cosine,

$$V_d = V_{d0}\cos\alpha = V_{d0}\Big(\frac{2U_{cm}}{U_h} - 1\Big),$$

so that **the average output voltage becomes linear in the control** U_{cm}. The mapping is $U_{cm} = 0 \Rightarrow \alpha = 180^\circ \Rightarrow V_d = -V_{d0}$, $U_{cm} = 5 \Rightarrow \alpha = 90^\circ \Rightarrow V_d = 0$, and $U_{cm} = 10 \Rightarrow \alpha = 0^\circ \Rightarrow V_d = +V_{d0}$. The synchronised *Pulse Generator* (*6-pulse, 60° pulse width*) then places the six gate pulses at that angle relative to the measured mains phase.

2.4 Averaged converter model — the $22/(T_cs+1)$ block, and why a transfer function

For control design the switching bridge is unusable directly: it is **non-linear** (before linearisation) and **discrete** (it can only act at the six firing instants per period), whereas pole–zero and settling-time design needs a *linear, continuous* plant. The standard remedy is the **average-value model**, which replaces the whole converter — bridge, firing law and pulse generator — by two simple effects:

- **a static gain** K_c from control to average voltage. Thanks to the arccos law of Section 2.3 this gain is *constant* (the converter has been linearised). In the idealised block-diagram model the control range $U_{cm} \in [0, 10]$ is scaled so that full control produces the **rated armature voltage**:

$$K_c = \frac{U_{a,\text{rated}}}{U_{cm,\max}} = \frac{220}{10} = 22 \quad [\text{V per control unit}].$$

- **a small transport lag** T_c. After the controller changes U_{cm}, the new firing angle can only take effect at the *next* thyristor firing. Firings occur every $1/(6\times 50) = 3.33$ ms, so the average delay is half of that:

$$T_c = \frac{1}{2\times 6\times 50} = 1.667 \text{ ms},$$

  modelled as a first-order lag $1/(T_cs + 1)$.

Combining the two, the rectifier is represented by the single first-order block that appears in the Simulink block diagram (numerator 22, denominator $T_cs + 1$):

$$\frac{U_a(s)}{U_{cm}(s)} = \frac{K_c}{T_cs+1} = \frac{22}{T_cs+1}.$$

Why a transfer function rather than the switching bridge? Because it turns the converter into a linear element that can be folded into the loop transfer function, so the current PID can be tuned in closed form (Section 4). The switching model is kept separately (Section 6) to *verify* that the controllers designed on this average model still behave correctly once the real ripple, dead-time and discrete firing are present. This static gain also sets the current-loop plant gain used in the tuning,

$$b = \frac{K_c}{R_a} = \frac{22}{2.33} = 9.442 \quad [\text{A per control unit}],$$

i.e. control $\to$ voltage ($\times 22$) $\to$ current ($\times 1/R_a$).

3. Plant transfer function and its Simulink realisation

3.1 Deriving $G(s)$

Taking Laplace transforms of the two balances of Section 1.1 with the load written as a viscous term ($C_r = K_r\omega$):

$$U_a = (R_a + (L_a + L_f)s)I_a + k\Phi\,\Omega, \qquad J s\,\Omega = k\Phi\, I_a - (f_r + K_r)\Omega.$$

Eliminating I_a gives

$$G(s) = \frac{\Omega}{U_a} = \frac{k\Phi}{(R_a + (L_a{+}L_f)s)(Js + f_r + K_r) + k\Phi^2}.$$

3.2 The *Moteur DC* block

The user-built *Moteur DC* subsystem realises exactly this transfer function as an explicit block diagram (Figure 2), which makes the internal current and speed signals available for measurement and for the inner loop.

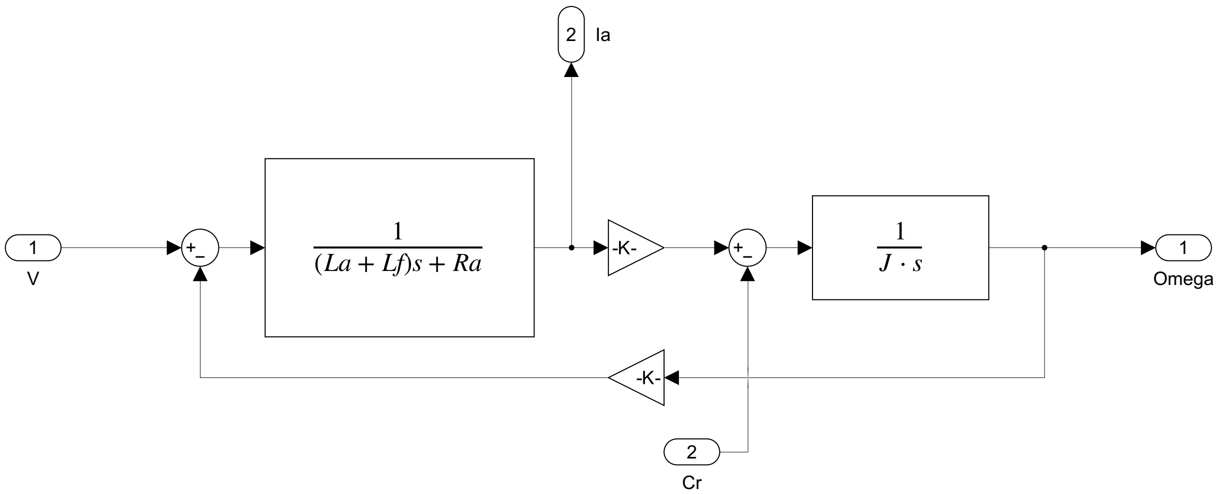


Figure 2. The *Moteur DC* subsystem: the armature admittance $1/((L_a{+}L_f)s + R_a)$ produces the current, $k\Phi$ converts it to torque, $1/(Js)$ integrates to speed, and $k\Phi\,\omega$ is fed back as the back-EMF.

Following the diagram, the armature voltage V has the back-EMF $k\Phi\,\omega$ subtracted, drives the armature admittance to give the current I_a, which is scaled by $k\Phi$ into torque; the load torque C_r is subtracted and the net torque is integrated by $1/(Js)$ into the speed Ω:

$$I_a = \frac{V - k\Phi\,\Omega}{(L_a + L_f)s + R_a}, \qquad \Omega = \frac{k\Phi\, I_a - C_r}{Js}.$$

Substituting one into the other returns $G(s)$ exactly, confirming that the block diagram and the analytical model agree.

3.3 Fixing the operating point

The load is not a free parameter: the viscous coefficient K_r is **back-computed** so that the requested operating point (220 V producing 3000 rpm) is exactly the machine's steady state,

$$K_r = \frac{1}{R_a}\left(\frac{k\Phi\, U_a}{\omega^\star} - k\Phi^2\right) - f_r.$$

With $U_a = 220$ V and $\omega^\star = 3000 \cdot 2\pi/60 = 314.16$ rad/s this gives $K_r = 0.03170$, a DC gain $G(0) = 1.42800\ \mathrm{rad\,s^{-1}\,V^{-1}}$, and hence $\omega_{ss} = G(0)\cdot 220 = 3000.0$ rpm — self-consistent. Two characteristic time constants follow and are used in the tuning:

$$\tau_e = \frac{L_a + L_f}{R_a} = 0.1230 \text{ s (electrical)}, \qquad \tau_m = \frac{J R_a}{(k\Phi)^2} = 0.0538 \text{ s (mechanical)}.$$

4. Controller design

Both controllers are the Simscape *PID (2-DOF disabled, parallel form)* block, $C(s) = K_p + K_i/s + K_d \frac{N}{1+N/s}$ with derivative-filter coefficient $N = 100$; the speed loop uses only the PI terms. They are tuned analytically with two ideas: **pole–zero cancellation** (place the controller zeros on the plant poles so the open loop becomes a pure integrator) and a **settling-time criterion** (a first-order lag of time constant τ has a 1 % settling time $t_s = -\ln(0.01)\,\tau \approx 4.6\,\tau$, the source of the recurring factor 4.6).

4.1 Inner current loop (PID)

Neglecting the slow back-EMF coupling, the plant from control to current is the converter lag in series with the armature electrical lag,

$$P_i(s) = \frac{b}{(1+\tau_e s)(1+T_c s)}, \qquad b = \frac{22}{R_a} = 9.442.$$

Writing the PID with its zeros exposed and choosing $K_p/K_i = \tau_e + T_c$ and $K_d/K_i = \tau_e T_c$ makes its numerator factor as $(1+\tau_e s)(1+T_c s)$, cancelling both plant poles. The open loop collapses to $L(s) = K_i b/s$ and the closed loop is first-order with $\tau_{\text{cl}} = 1/(K_i b)$; imposing $t_s = 4.6\,\tau_{\text{cl}} = t_{r,i}$ fixes the gains:

$$K_i = \frac{4.6}{t_{r,i}\,b}, \qquad K_p = (\tau_e + T_c)K_i, \qquad K_d = \tau_e T_c\,K_i.$$

For $t_{r,i} = 0.1$ s: $K_i = 4.872$, $K_p = 0.607$, $K_d = 0.001$. The derivative term is tiny because T_c is tiny, so the controller is essentially a PI with a small correction for the converter dead-time.

4.2 Outer speed loop (PI)

With the current loop fast and near-unity in its bandwidth, the mechanical plant seen by the speed controller reduces to a first-order lag between current demand and speed, $P_\omega(s) = a/(1 + \frac{J}{k}s)$ with $a = k\Phi/k = 45.6$. Here $k = 0.0125$ plays the role of an equivalent damping/measurement scaling. A PI controller cancels the mechanical pole when $K_p/K_i = J/k$, again reducing the open loop to $K_i a/s$; imposing $t_s = 4.6/(K_i a) = t_{r,w}$ gives

$$K_i = \frac{4.6}{t_{r,w}\,a}, \qquad K_p = \frac{J}{k}\,K_i.$$

For $t_{r,w} = 1$ s: $K_i = 0.1009$, $K_p = 0.0605$.

4.3 Summary and unit handling

The tuned gains are collected in Table 2. In the models, speed is measured through a scaling gain $k = 1.25 \times 10^{-2}$ and the reference/display use the unit conversions $60/2\pi$ (rad/s $\to$ rpm) and $2\pi/60$ (rpm $\to$ rad/s); the current reference is hard-limited by a saturation block to enforce current limiting.

Table 2. Analytically tuned controller gains.

Loop	Type	K_p	K_i	K_d	Target response
Current (inner)	PID	0.607	4.872	0.001	$t_{r,i} = 0.1$ s
Speed (outer)	PI	0.0605	0.1009	—	$t_{r,w} = 1$ s

4.4 Alternative tuning for a constant load torque

For a constant load torque the speed loop can instead be tuned by direct **second-order pole placement** with a chosen damping ratio ζ and natural frequency ω_n:

$$K_p = \frac{2\zeta\omega_n J}{k\Phi}, \qquad K_i = \frac{\omega_n^2 J}{k\Phi},$$

with the current loop using $\tau_m = 53.8$ ms in place of $\tau_e + T_c$. This variant is analysed in Section 5 because it makes the roles of ζ and ω_n directly visible.

5. Python simulation results

The MATLAB design was re-implemented independently in Python (NumPy, SciPy, Matplotlib) to confirm the tuning and to visualise how the closed-loop response depends on the design targets t_r, ζ and ω_n.

5.1 Cross-check against MATLAB

The Python port reproduces every design quantity to the printed precision (Table 3).

Table 3. Design quantities: MATLAB versus the independent Python port.

Quantity	MATLAB	Python
Load coefficient K_r	0.031697	0.031697
DC gain $G(0)$	1.42800	1.42800
Steady speed	3000.0 rpm	3000.0 rpm
Current PID $K_p/K_i/K_d$	0.607/4.872/0.001	0.607/4.872/0.001
Speed PI K_p/K_i	0.0605/0.1009	0.0605/0.1009

At the designed operating point the two loops give the first-order responses of Figure 3, showing the deliberate ten-to-one time-scale separation (current ≈ 0.1 s, speed ≈ 1 s).

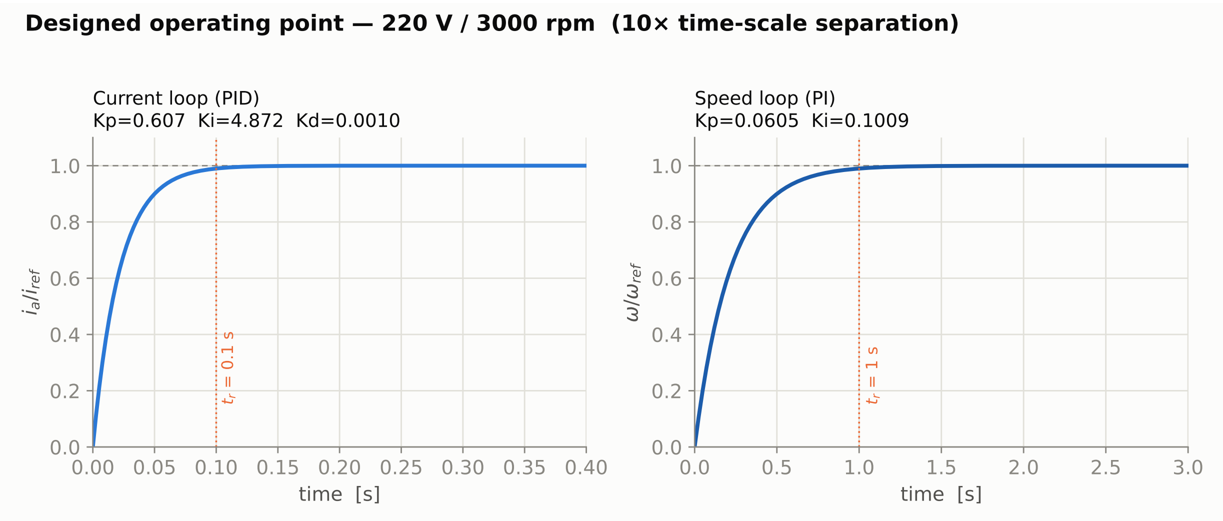


Figure 3. Designed operating point: step responses of the inner current loop (left) and outer speed loop (right).

5.2 Order 1 — the response time sets the rise time

After pole–zero cancellation the current loop is exactly first-order, $T(s) = 1/(1 + \tau s)$ with $\tau = t_{r,i}/4.6$. Sweeping $t_{r,i}$ slides the single pole and hence the rise time, with no overshoot at any setting (Figure 4, Table 4).

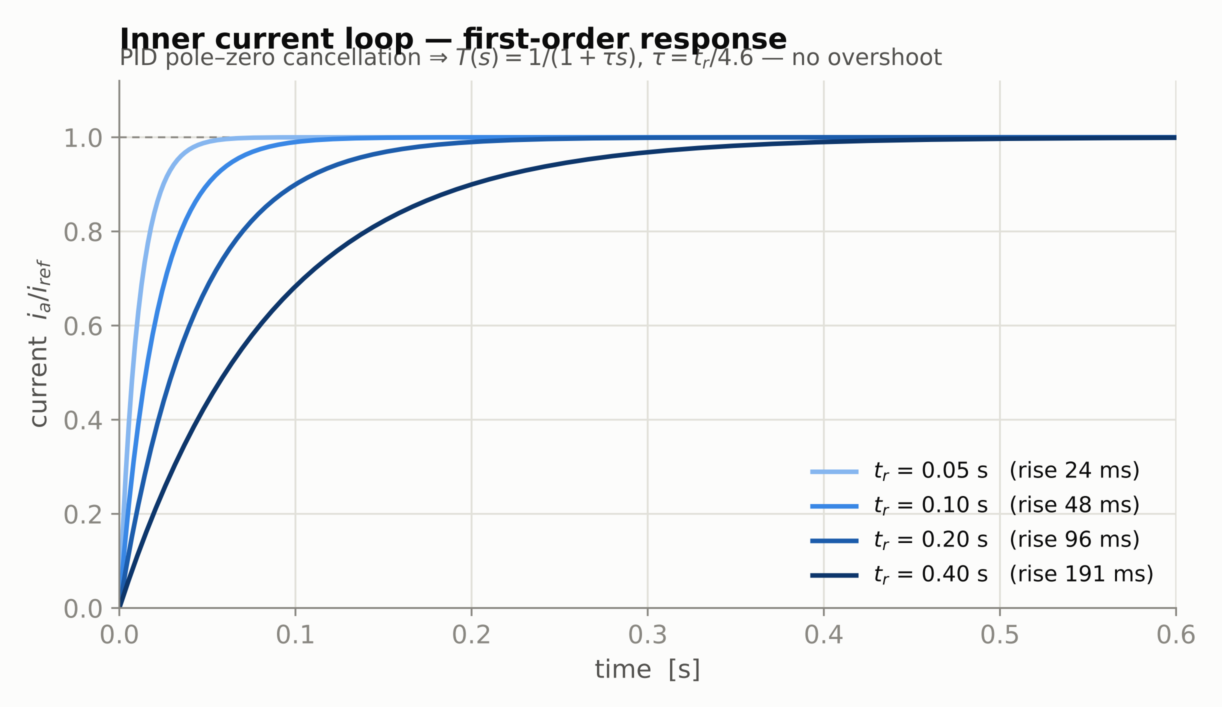


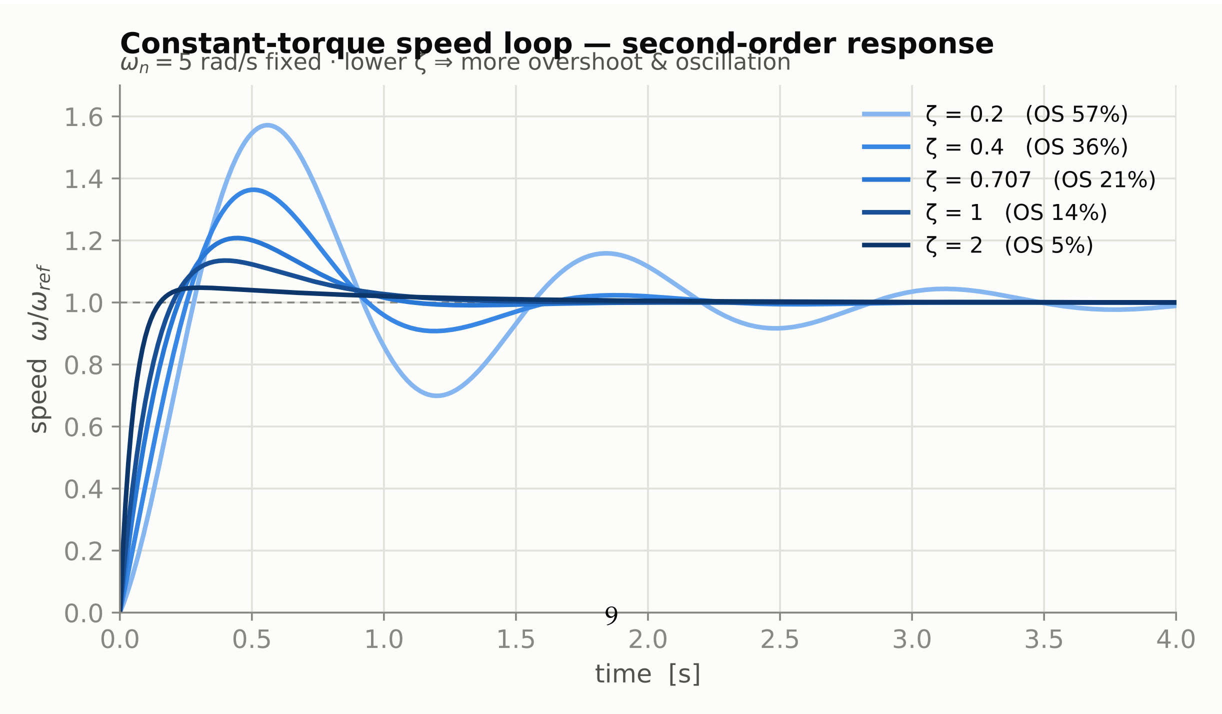
Figure 4. First-order current loop: the response-time target sets the rise time, with no overshoot.

Table 4. First-order current loop: rise time versus the response-time target.

$t_{r,i}$	closed-loop τ	10–90 % rise time
0.05 s	10.9 ms	24 ms
0.10 s	21.7 ms	48 ms
0.20 s	43.5 ms	96 ms
0.40 s	87.0 ms	191 ms

5.3 Order 2 — the damping ratio sets the oscillation

For the constant-torque design the speed loop is second-order, $T(s) = (2\zeta\omega_n s + \omega_n^2)/(s^2 + 2\zeta\omega_n s + \omega_n^2)$. Holding $\omega_n = 5$ rad/s and sweeping ζ moves the poles from lightly damped to overdamped (Figure 5, Table 5). The overshoot is higher than the textbook value (21 % rather than $~$4 % at $\zeta = 0.707$) because the PI controller adds a finite zero at $s = -\omega_n/(2\zeta)$, which speeds the rise but amplifies the first peak — precisely why the nominal design prefers pole–zero cancellation.



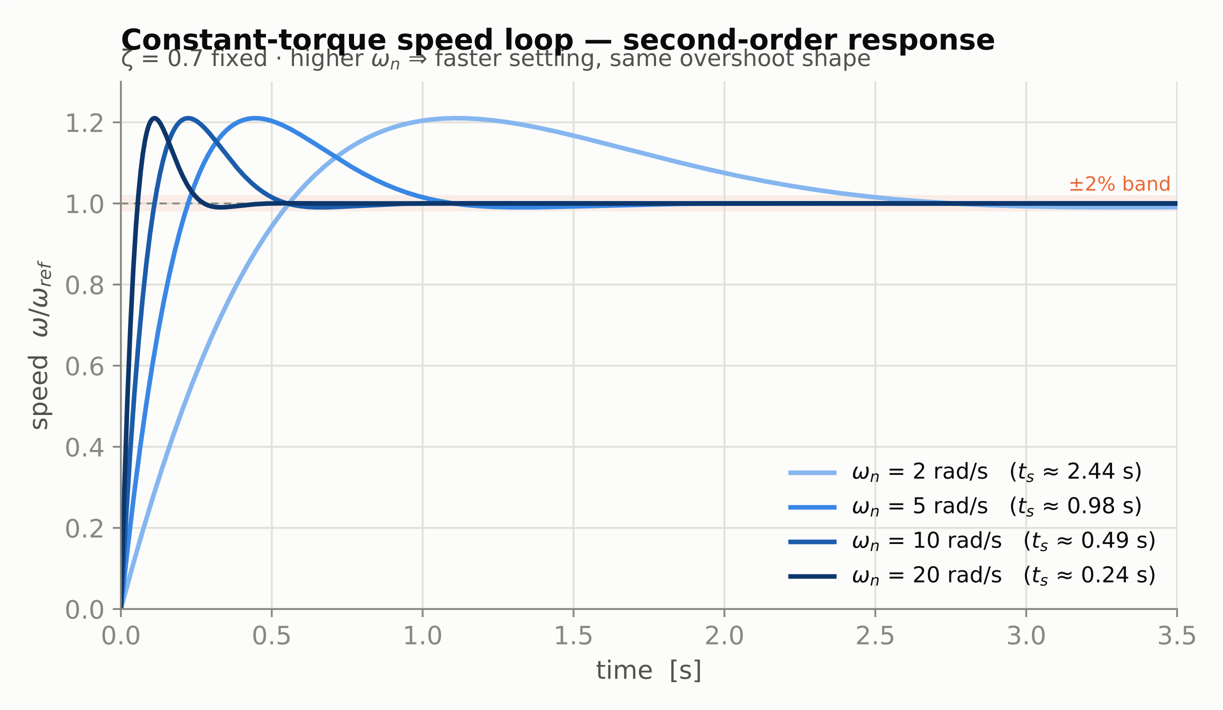


Figure 6. Second-order speed loop: the natural frequency sets the settling time ($\zeta = 0.7$ fixed).

Table 6. Second-order speed loop: settling time versus natural frequency.

ω_n	$\pm 2\ \%$ settling time
2 rad/s	2.44 s
5 rad/s	0.98 s
10 rad/s	0.49 s
20 rad/s	0.24 s

6. Simulink model results

Three complementary models are used, in increasing order of physical fidelity; the diagrams are exported directly from the models and run on a continuous `powergui` solver.

6.1 Transfer-function block diagram

The block-diagram model (Figure 7) is the idealised realisation that matches Sections 2–4 one-to-one: the averaged converter is the $22/(T_c s + 1)$ block, the machine is the *Moteur DC* subsystem of Figure 2, and the compensators are the inner PID and outer PI. Because this *is* the design model, its response is the reference: the current settles in ≈ 0.1 s and the speed reaches 3000 rpm in ≈ 1 s with little overshoot.

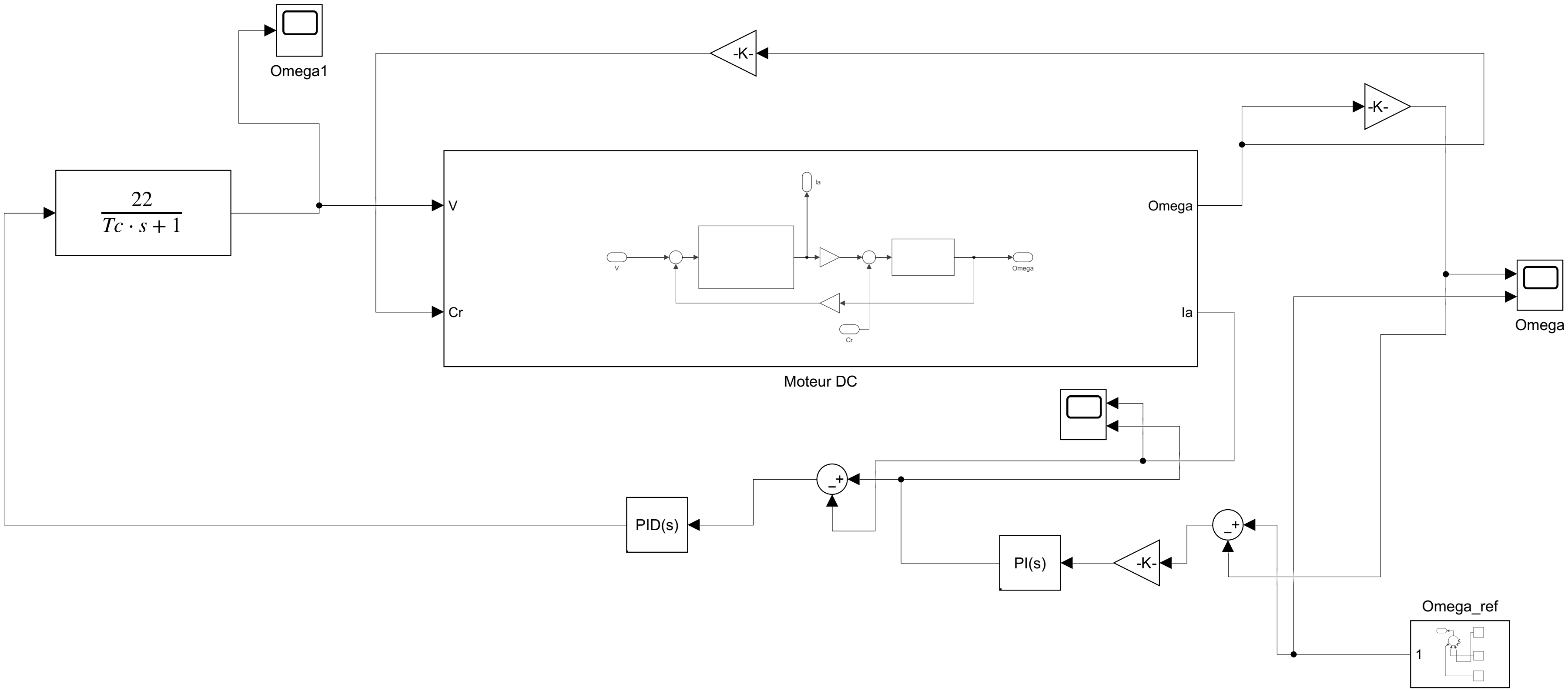


Figure 7. Transfer-function cascade block diagram (the idealised design model).

6.2 Physical model with the detailed thyristor rectifier

Figure 8 is the highest-fidelity model, built from Simscape Electrical library blocks: the three-phase source, the six-pulse thyristor bridge, the *Alpha generator* (Figure 1) and synchronised pulse generator, the 276 mH smoothing inductor, and the *DC Machine* block, with the same cascade controllers. Relative to the averaged model it exposes exactly what the tuning was designed to tolerate: the 300 Hz current ripple, the firing dead-time $T_c = 1.667$ ms embedded in the current-loop gains, and the discrete nature of thyristor conduction. The dead-time term in the PID and the current-reference saturation keep the drive stable and current-limited despite the ripple, and the mean current and speed still track the 3000 rpm target.

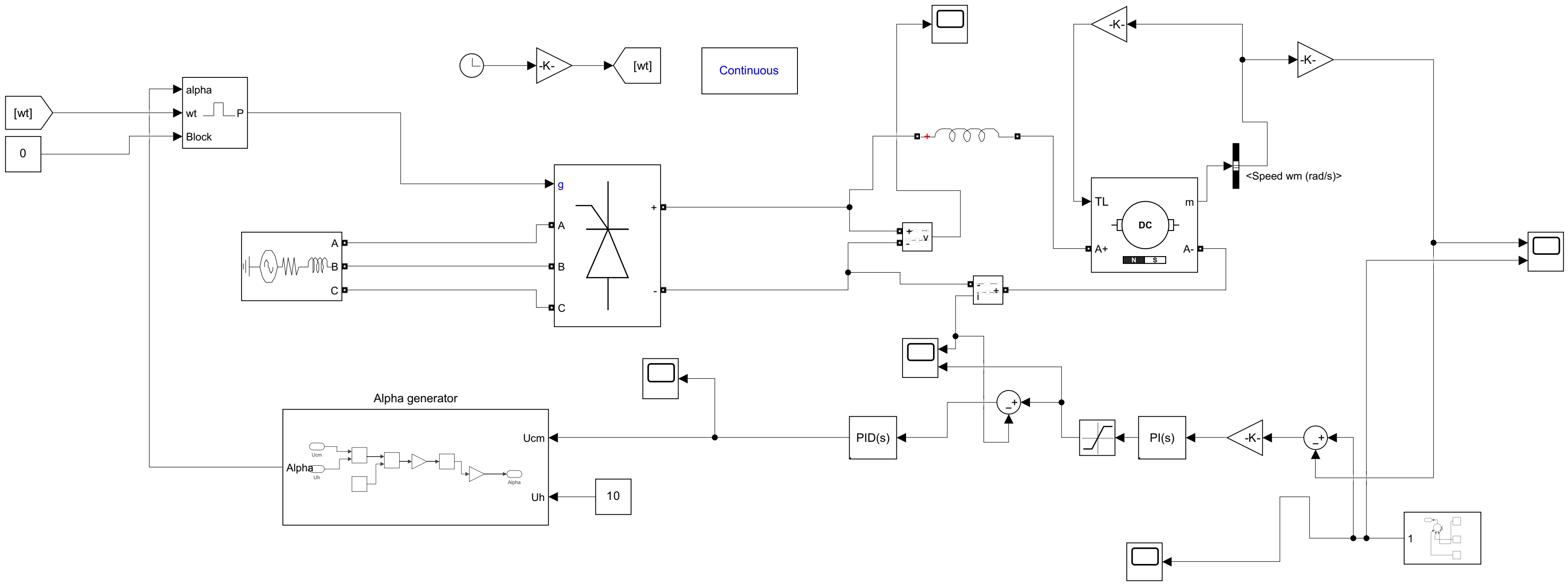


Figure 8. Detailed physical model: three-phase source, six-pulse thyristor bridge, Alpha generator, smoothing inductor and DC machine.

6.3 Physical model with an averaged source

Figure 9 is an intermediate model that keeps the physical *DC Machine* block — so the electrical and mechanical dynamics are the genuine Simscape ones — but drives it from a controlled voltage source rather than the full bridge (its gain $\frac{3\sqrt{6}}{\pi} \cdot \frac{220}{10}$ is the same averaged-converter idea as the 22 of Section 2.4). It isolates the machine physics from the converter physics and is the natural stepping-stone between the idealised block diagram and the detailed rectifier model.

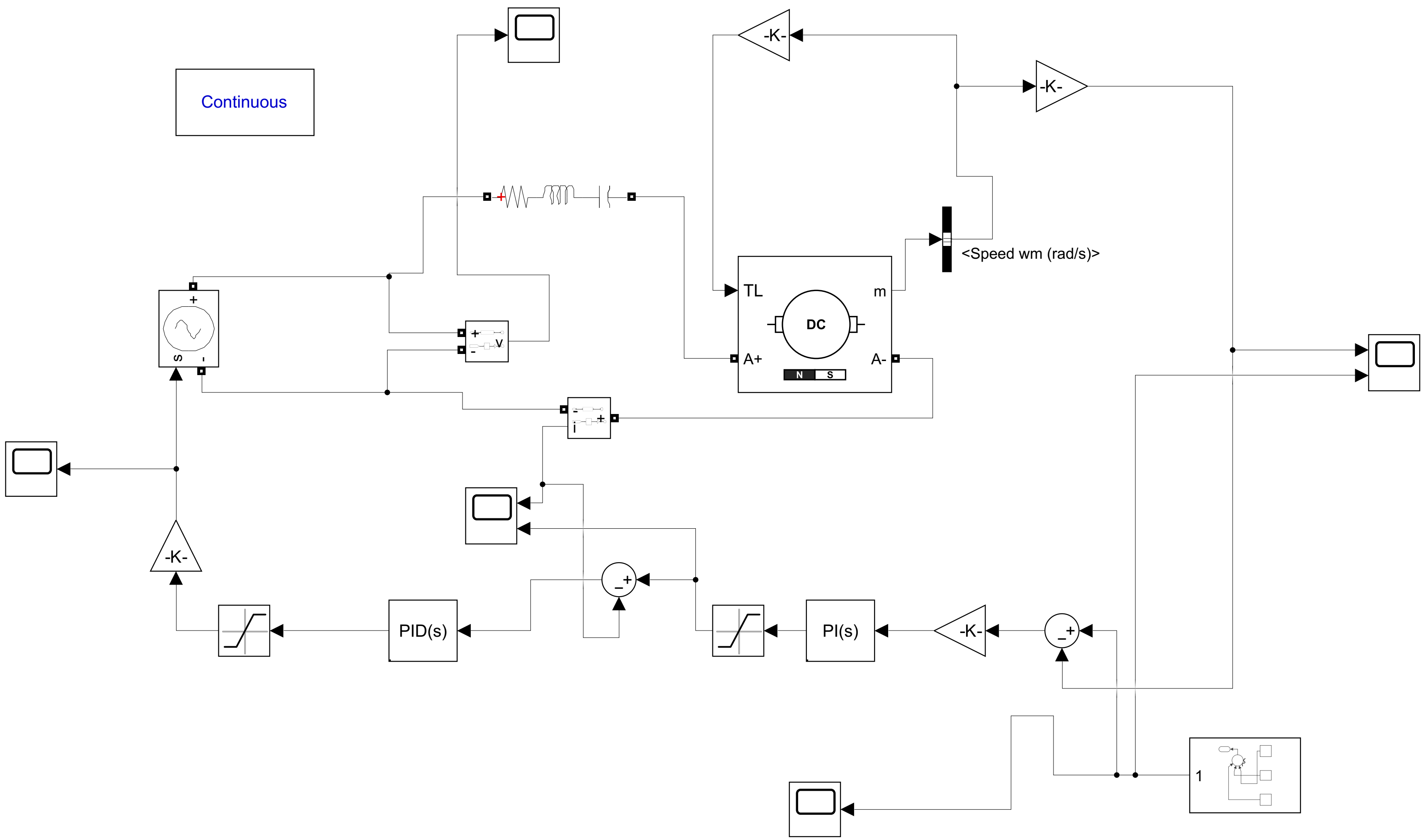


Figure 9. Intermediate physical model: genuine DC-machine block driven by an averaged controlled source.

7. Conclusion

A cascade speed-and-current controller for a thyristor-fed separately-excited DC machine was developed in full detail. The controlled rectifier was analysed from first principles — the six-pulse bridge, its cosine voltage law, and the *Alpha generator* whose arccos firing law linearises the converter — and then condensed into the averaged model $22/(T_c s + 1)$, whose gain $(22 = 220/10)$ and lag $(T_c = 1.667 \text{ ms})$ were derived from the drive's own quantities. On that linear plant the PI and PID gains were obtained analytically by pole–zero cancellation and a settling-time criterion, giving transparent, independently adjustable design targets. The design was verified twice: a Python model reproduced the tuning exactly and quantified the parameter dependencies, and a hierarchy of MATLAB/Simulink models — ending with the full switching bridge — confirmed the intended behaviour at the 220 V / 3000 rpm operating point.